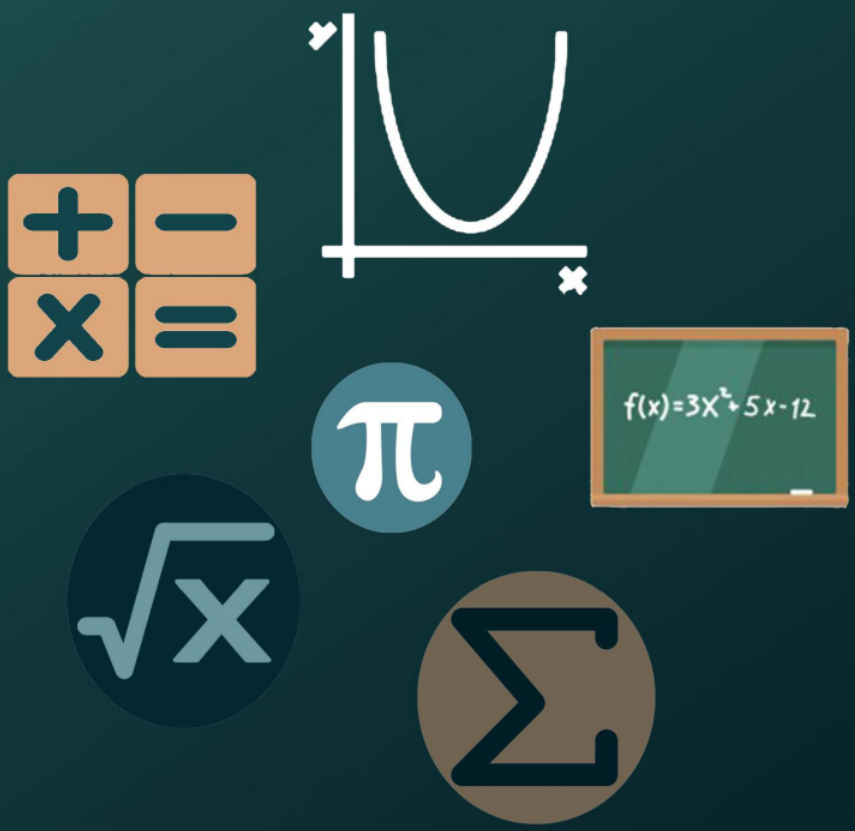
MATHEMATICS

Statistics



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No	Topic Name	Ex 1	Ex 2
1	Measure of Central Tendency - Arithemetic Mean	1 - 6	1
2	Properties of Arithemetic mean		2, 3
3	Weighted Mean and Combined Mean		4, 5
4	Harmonic Mean and Geometric Mean	7,8	
5	Measure of Central Tendency - Median	9, 10	6
6	Measure of Central Tendency - Mode	11 - 13	
7	Measures of Dispersion - Range and Mean Deviation	14 - 17	7 - 9
8	Measures of Dispersion - Variance and Standard Deviation	18 - 23	10 - 19
9	Properties of Variance	24	20

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Arithmetic Mean

Q1. If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to-

(A) 10 (B) 11

(C) 12 (D) none of these

Q2. The mean of values $1, \frac{1}{2}, \frac{1}{3}, \ldots\ldots \frac{1}{n}$ which have frequencies 1, 2, 3, ... n resp., is :-

(A) $\frac{2n+1}{3}$ (B) $\frac{2}{n}$

(C) $\frac{n+1}{2}$ (D) $\frac{2}{n+1}$

Q3. In a group of students, the mean weight of boys is 65 kg. and mean weight of girls is 55 kg. If the mean weight of all students of group is 61 kg, then the ratio of the number of boys and girls in the group is :-

(A) 2 : 3 (B) 3 : 1

(C) 3 : 2 (D) 4 : 3

Q4. If the mean of 3, 4, x, 7, 10 is 6, then the value of x is

(A) 4 (B) 5

(C) 6 (D) 7

Q5. Consider the frequency distribution of the given numbers

Value :	1	2	3	4
Frequency :	5	4	6	f

If the mean is known to be 3, then the value of f is :

(A) 3 (B) 7

(C) 10 (D) 14

Q6. The number of observations in a group is 40. If the average of first 10 is 4.5 and that of the remaining 30 is 3.5, then the average of the whole group is :

(A) $\frac{1}{5}$ (B) $\frac{15}{4}$ (C) 4 (D) 8

Harmonic mean and Geometric Mean

Q7. The G.M. of numbers 4, 5, 10, 20, 25 is :-

(A) 12.8 (B) 10

(C) 7.8 (D) None of these

Q8. The H.M. of the numbers 2, 3, 4 is :-

(A) 3 (B) $2(3)^{1/3}$

(C) $\frac{36}{13}$ (D) $\frac{13}{36}$

Measure of central Tendency-Median

Q9. Median of the following freq. dist.

x_i	3	6	10	12	7	15
f_i	3	4	2	8	13	10

(A) 7 (B) 10

(C) 8.5 (D) None of these

Q10. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observation of the set is increased by 2, then the median of the new set :

(A) Is increased by 2

(B) Is decreased by 2

(C) Is two times the original median

(D) Remains the same as that of the original set

Measure of central Tendency-Mode

Q11. The mode of the following freq. dist is :-

Class	1–10	11–20	21–30	31–40	41–50
f_i	5	7	8	6	4

(A) 24 (B) 23.83
(C) 27.16 (D) None of these

Q12. A set of numbers consists of three 4's, five 5's, six 6's, eight 8's and seven 10's. The mode of this set of numbers is :
(A) 6 (B) 7
(C) 8 (D) 10

Q13. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately.
(A) 25.5 (B) 24.0
(C) 22.0 (D) 20.5

Measure of Dispersion - Range and Mean Deviation

Q14. Mean deviation of the observations 70, 42, 63, 34, 44, 54, 55, 46, 38, 48 from median is :-
(A) 7.8 (B) 8.6
(C) 7.6 (D) 8.8

Q15. Which of the following is not a measure of central tendency.
(A) Mean (B) Median
(C) Mode (D) Range

Q16. The mean deviation from the mean for the set of observations – 1,0,4 is
(A) $\sqrt{\frac{14}{3}}$ (B) 2
(C) $\frac{2}{3}$ (D) None of these

Q17. A batsman scores runs in 10 innings 38, 70, 48, 34, 42, 55, 63, 46, 54, 44, then the mean deviation is -
(A) 8.6 (B) 6.4
(C) 10.6 (D) 9.6

Measure of Dispersion - Variance and Standard Deviation

Q18. If $\sum_{i=1}^{10}(x_i-15)=12$ and $\sum_{i=1}^{10}(x_i-15)^2=18$ then the S.D. of observations $x_1, x_2 \ldots\ldots x_{10}$ is :-
(A) $\frac{2}{5}$ (B) $\frac{3}{5}$
(C) $\frac{4}{5}$ (D) None of these

Q19. The variance of series a, a + d, a + 2d, ..., a + 2nd is :-
(A) $\frac{n(n+1)}{2}d^2$ (B) $\frac{n(n+1)}{3}d^2$
(C) $\frac{n(n+1)}{6}d^2$ (D) $\frac{n(n+1)}{12}d^2$

Q20. The variance of the first n natural numbers is :-
(A) $\frac{n^2-1}{12}$ (B) $\frac{n^2-1}{6}$
(C) $\frac{n^2+1}{6}$ (D) $\frac{n^2+1}{12}$

Q21. If v is the variance and σ is the standard deviation, then
(A) $v^2=\sigma$ (B) $v=\sigma^2$
(C) $v=\frac{1}{\sigma}$ (D) $v=\frac{1}{\sigma^2}$

Q22. The mean and S.D. of 1,2,3,4,5,6 is

(A) $\frac{7}{2}, \sqrt{\frac{35}{12}}$ (B) 3,3

(C) $\frac{7}{2}, \sqrt{3}$ (D) $3, \frac{35}{12}$

Q23. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The co-efficient of variation is :-

(A) 50% (B) 10%

(C) 40% (D) None of these

Properties of Variance

Q24. The variance of α, β and γ is 9, then variance of $5\alpha, 5\beta$ and 5γ is

(A) 45 (B) $\frac{9}{5}$

(C) $\frac{5}{9}$ (D) 225

Exercise - 2 Brain Booster

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In a class of 50 students, 10 have failed and their average marks are 28. The total marks obtained by the entire class are 2800. The average marks of those who have passed, are :

(A) 43 (B) 53

(C) 63 (D) 70

Q2. If the mean of five observations x, $x+2$, $x+4$, $x+6$ and $x+8$ is 11, then the mean of last three observations is :

(A) 13 (B) 15

(C) 17 (D) None of these

Q3. The mean of a set of observations is $\bar{x}$. If each observation is divided by α, $(\alpha \neq 0)$ and then is increased by 10, then the mean of the new set is :

(A) $\frac{\bar{x}}{\alpha}$ (B) $\frac{\bar{x}+10}{\alpha}$

(C) $\frac{\bar{x}+10\alpha}{\alpha}$ (D) $\frac{\alpha\bar{x}+10}{\alpha}$

Q4. The mean of values 0, 1, 2, ..., n when their weights are $1, {}^nC_1, {}^nC_2, ..., {}^nC_n$, resp., is :

(A) $\frac{2^n}{n+1}$ (B) $\frac{n+1}{2}$

(C) $\frac{2^{n+1}}{n(n+1)}$ (D) $\frac{n}{2}$

Q5. The average marks of boys in a class 52 and that of girls is 42. The average marks of boys and girls combined is 50 then the parcentage of boys in the class is -

(A) 20 (B) 80

(C) 60 (D) 40

Q6. If a variable takes the discrete values $\alpha+4$, $\alpha-\frac{7}{2}, \alpha-\frac{5}{2}, \alpha-3, \alpha-2, \alpha+\frac{1}{2}, \alpha-\frac{1}{2}$, $\alpha+5(\alpha>0)$, then the median is :

(A) $\alpha-\frac{5}{4}$ (B) $\alpha-\frac{1}{2}$

(C) $\alpha-2$ (D) $\alpha+\frac{5}{4}$

Q7. The mean deviation of the data 3, 10, 10, 4, 7, 10, 5 from the mean is

(A) 2 (B) 2.57

(C) 3 (D) 3.75

Q8. When tested, the lives (in hours) of 5 bulbs were noted as follows :

1357, 1090, 1666, 1494, 1623

The mean deviations (in hours) from their mean is -

(A) 178 (B) 179

(C) 220 (D) 356

Q9. Following are the marks obtained by 9 students in a Mathematics test :

50, 69, 20, 33, 53, 39, 40, 65, 59

The mean deviation from the median is -

(A) 9 (B) 10.5

(C) 12.67 (D) 14.76

Q10. The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is -

(A) $\sqrt{\frac{52}{7}}$ (B) $\frac{52}{7}$

(C) $\sqrt{6}$ (D) 6

Q11. The mean of 100 observations is 50 and their standard deviation is 5. The sum of all squares of all the observations is -

(A) 50000 (B) 250000

(C) 252500 (D) 255000

Q12. The SD of 15 items is 6 and if each item is decreases by 1, then standard deviation will be :

(A) 5 (B) 7

(C) $\frac{91}{15}$ (D) 6

Q13. Standard deviations for first 10 natural numbers is -

(A) 5.5 (B) 3.87

(C) 2.97 (D) 2.87

Q14. If $\sum_{i=1}^{18}(x_i-8)=9$ and $\sum_{i=1}^{18}(x_i-8)^2=45$, then the standard deviation of $x_1, x_2, \ldots. x_{18}$ is -

(A) $\frac{4}{9}$ (B) $\frac{9}{4}$

(C) $\frac{3}{2}$ (D) None of these

Q15. If the standard deviation of $x_1, x_2, \ldots\ldots, x_n$ is 3.5, then the standard deviation of $-2x_1-3, -2x_2-3, \ldots.., -2x_n-3$ is -

(A) –7 (B) –4

(C) 7 (D) 1.75

Q16. The first of two samples has 100 items with mean 15 sand SD = 3. If the whole group has 250 items with mean 15.6 and $SD=\sqrt{13.44}$, the SD of the second group is -

(A) 4 (B) 5

(C) 6 (D) 3.52

Q17. In a series of 2n observations, half of them equal a and remaining half equal –a. If the standard deviation of the observations is 2, then |a| equals :

(A) 2 (B) $\sqrt{2}$

(C) $\frac{1}{n}$ (D) $\frac{\sqrt{2}}{n}$

Q18. The standard deviation of variate x_i is s. Then standard deviation of the variate $\frac{ax_i+b}{c}$, where a, b, c are constants is -

(A) $\left(\frac{a}{c}\right)\sigma$ (B) $\left|\frac{a}{c}\right|\sigma$

(C) $\left(\frac{a^2}{c^2}\right)\sigma$ (D) None of these

Q19. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80 then which one of the following gives possible values of a and b ?

(A) a = 0, b = 7

(B) a = 5, b = 2

(C) a = 1, b = 6

(D) a = 3, b = 4

Q20. Suppose a population A has 100 observations 101, 102, ... 200 and other population B has 100 observations 151, 152, ... 250. If V_A and V_B represent the variance of two population respectively then $\frac{V_A}{V_B}$ is-

(A) 9/4 (B) 4/9

(C) 2/3 (D) 1

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	D	C	C	D	B	B	C	C	D
11	12	13	14	15	16	17	18	19	20
B	C	B	B	D	B	A	B	B	A
21	22	23	24						
B	A	B	D						

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	A	C	D	B	A	B	A	C	A
11	12	13	14	15	16	17	18	19	20
C	D	D	C	C	A	A	B	D	D

-: Important Notes & Formulas :-